9	**(a)**		eg randomly select N different businesses and then randomly select P computers from each business; may be implied by correct description	**B1** **[1]**	**2.4**	$N \times P = 120$ where N and P are integers greater than 1.	eg 20 and 6 or 15 and 8

Question			Answer	Marks	AOs		Guidance
9	(b)		H_0: $\mu = 5$ **oe** H_1: $\mu \neq 5$ **oe**	**B1**	**1.1**	allow any parameter apart from $\bar{x}$ for population mean as long as clearly defined as (population) mean	after **B** marks **M1A1M1A1A1** may be earned if working with 2.7054 rounded to 2 or more sf
			H_1 takes this form as Claud is testing whether the mean length of time is different to 5 **oe**	**B1**	**2.4**		
			μ is the **population** mean time for which computers are kept before being replaced	**B1**	**2.5**		
			use of N(5, $\frac{2.7054^2}{120}$) to find P($\bar{X} < 4.8855$) or invNorm(p,5, $\frac{2.7054}{\sqrt{120}}$) where p = 0.025 or 0.05; may be implied by 0.3215 or 4.51595…or 4.59377…	**M1**	**3.3**	condone use of 2.6941 (may be rounded) instead of 2.7054 for **M1**, may be implied by 0.32076 or CR is $\bar{X} < 4.51797..$, but **A1** not available	or $z = \frac{4.8855 - 5}{\frac{2.7054}{\sqrt{120}}}$ for **M1** awrt − 0.46 **A1** (may be implied by –0.466 or –0.465567.. if 2.6941 used)
			P($\bar{X}$< 4.8855) = awrt 0.32	**A1**	**1.1**	or CR is $(\bar{X}) < 4.5 - 4.6$	
			0.32 > 0.025 or 4.8855 > 4.5(2)	**M1**	**3.4**	comparison of *their* probability with 0.025 **or** comparison of 4.8855 with *their* critical value from use of 0.025, as long as previous **M1** awarded	*their* − 0.4436 > – 1.96 **oe** **M1** dep on award of previous **M1**
			not significant or accept H_0 or do not reject H_0 or reject H_1	**A1FT**	**1.1**	may be embedded in conclusion in context	
			insufficient evidence to **suggest** (at 5% level) that the (population) **mean** length of time (computers are kept) is not 5 years	**A1FT** **[8]**	**2.2b**		do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; **A0** if answer spoiled